

Curve-Warp AMMs:

A Geometric Formalization of Weight-Adjusted Liquidity Provision

Temporal Exchange

Abstract

We formalize a class of weighted-geometric AMMs in which trade impact is realized by deforming the pricing curve about a fixed *anchor point* on the curve, while the trade pivot is permitted to slide radially along its polar ray onto the new curve. In log-log coordinates the pricing curve becomes a straight line and the deformation reduces to a rotation about the anchor. The new tangent at the trade ray is set equal to the tangent slope at the would-be endpoint of the trader’s walk along the pre-trade curve—a marginal-price target, not a secant. We identify the natural intrinsic coordinate—the *rapidity* $\xi = \ln \tan \theta$ —and prove a local conservation law: the log-slope is affine in rapidity with coefficient exactly 1 regardless of pool weight. Strikes are characterized by trade-invariant rapidities; the pool’s mode rapidity shifts under a trade by exactly the negative of the walked rapidity, $\Delta \xi_m = \xi_* - \xi_B$. Reciprocal-strike pairs carry identical option premia independently of weight. Trades that move the mode past a strike’s rapidity flip the strike’s type (call $\leftrightarrow$ put).

1 Setup

1.1 The weighted-geometric curve

The pricing curve of a pool with state (w, k) , $w \in (0, 1)$, $k > 0$, is

$$V(x, y) := x^w y^{1-w} = k, \quad x, y > 0. \quad (1)$$

1.2 Polar coordinates

Parametrize the curve by $\theta \in (0, \pi/2)$ with $\tan \theta = y/x$. Eq. (1) gives

$$x(\theta) = k (\tan \theta)^{w-1}, \quad y(\theta) = k (\tan \theta)^w. \quad (2)$$

1.3 Slope and option premia

Differentiating (1) along the curve,

$$\frac{dy}{dx} = -\frac{w}{1-w} \cdot \frac{y}{x} = -\frac{w}{1-w} \tan \theta. \quad (3)$$

Write $\sigma(\theta) := \frac{w}{1-w} \tan \theta$ for the slope magnitude. The two option premia are read off as side-specific secant ratios:

$$P_C(\theta) := -\frac{dx}{dy} = \frac{\tan \theta_m}{\tan \theta} = \frac{1}{\sigma(\theta)}, \quad P_P(\theta) := -\frac{dy}{dx} = \frac{\tan \theta}{\tan \theta_m} = \sigma(\theta), \quad (4)$$

where the **mode angle** θ_m is the unique angle with $\sigma = 1$:

$$\tan \theta_m = \frac{1-w}{w}. \quad (5)$$

P_C is the call premium (notional in y , premium in x); P_P is the put premium (notional in x , premium in y , under the X -axis mirror convention). They satisfy the *premium duality*

$$P_C(\theta) \cdot P_P(\theta) \equiv 1, \quad (6)$$

and both equal 1 at the mode—the pool’s ATM (premium-neutral) point by construction.

1.4 Log-log linearization and rapidity

Let $u := \ln x$ and $v := \ln y$. Taking logs of (1),

$$w u + (1-w) v = \ln k. \quad (7)$$

Every pricing curve is a straight line in (u, v) space, with slope $-w/(1-w)$. The line intersects the diagonal $u = v$ at $(\ln k, \ln k)$; we call this the **diagonal-crossing**. The line also passes through the **ATM mode**—the point on the curve where $\sigma = 1$, with coordinates

$$(u_m, v_m) = (\ln k - (1-w) \xi_m, \ln k + w \xi_m), \quad (8)$$

so that $v_m - u_m = \xi_m$ (the mode rapidity). The diagonal-crossing and the mode coincide only when $w = 1/2$ (i.e. $\xi_m = 0$); for tilted pools they differ.

Define the **rapidity**

$$\xi := \ln \tan \theta = v - u, \quad \xi_m := \ln \tan \theta_m = \ln \frac{1-w}{w}. \quad (9)$$

The slope and premia rewrite as

$$\sigma(\xi) = e^{\xi - \xi_m}, \quad P_C(\xi) = e^{\xi_m - \xi}, \quad P_P(\xi) = e^{\xi - \xi_m}. \quad (10)$$

1.5 Strikes as trade-invariant rapidities

A strike is a point on the curve fixed in the (x, y) plane and is intrinsically labeled by either its polar angle θ_{strike} or its rapidity $\xi_{\text{strike}} = \ln \tan \theta_{\text{strike}}$. These are coordinate identifiers in the underlying plane and are therefore *invariant* under any trade. By contrast, the strike’s offset from the pool’s mode—measured either in percent (“+30% from spot”) or in any mode-relative angular form—is a *dynamic* reading that shifts whenever the trade moves θ_m .

Remark. The pool’s response to a trade—worked out in §5—reduces to a single scalar shift in ξ_m . Since premia at a strike depend only on $\xi_{\text{strike}} - \xi_m$, the entire effect of any trade on every strike on the book is captured by that one scalar.

2 Local invariants

Proposition 1 (Log-slope is affine in rapidity). *For any point on the curve (1),*

$$\ln \sigma(\xi) = \xi - \xi_m. \quad (11)$$

The relationship is affine in ξ with coefficient identically 1, independent of w and k .

Proof. From (3) and (9), $\sigma(\xi) = e^{-\xi_m} e^\xi$. □

Corollary 1 (Premia in rapidity). $\ln P_C(\xi) = -(\xi - \xi_m)$ and $\ln P_P(\xi) = \xi - \xi_m$; in particular $P_C \cdot P_P = 1$.

The pool weight w enters only through the origin ξ_m ; the pool scale k does not enter at all. In $(\xi, \ln \sigma)$ coordinates every pool is the same unit-slope line, translated.

3 The conservation law

Theorem 1 (Slope-product invariant). *Fix any point on the curve at rapidity ξ_* . For every $\Delta\xi$,*

$$\sigma(\xi_* + \Delta\xi) \cdot \sigma(\xi_* - \Delta\xi) = \sigma(\xi_*)^2. \quad (12)$$

Equivalently, $P_C(\xi_ + \Delta\xi) P_C(\xi_* - \Delta\xi) = P_C(\xi_*)^2$, and the same with P_P .*

Proof. By Proposition 1, $\ln \sigma(\xi_* \pm \Delta\xi) = (\xi_* \pm \Delta\xi) - \xi_m$. Summing yields $2(\xi_* - \xi_m) = 2 \ln \sigma(\xi_*)$. □

The local slope (and each premium) is the geometric mean of its equal-rapidity neighbors. The invariant holds at every point of every pool curve.

Corollary 2 (Hyperbolic structure of slope integrals). *Let $\Sigma_\pm(\Delta\xi) := \int_{\xi_*}^{\xi_* \pm \Delta\xi} \sigma d\xi$. Then*

$$\Sigma_+(\Delta\xi) + \Sigma_-(\Delta\xi) = 2 \sigma(\xi_*) \sinh(\Delta\xi), \quad (13)$$

$$\Sigma_+(\Delta\xi) \cdot \Sigma_-(\Delta\xi) = 2 \sigma(\xi_*)^2 (\cosh \Delta\xi - 1). \quad (14)$$

Proof. $\Sigma_+ = \sigma(\xi_*)(e^{\Delta\xi} - 1)$, $\Sigma_- = \sigma(\xi_*)(1 - e^{-\Delta\xi})$. Their sum is $\sigma(\xi_*)(e^{\Delta\xi} - e^{-\Delta\xi}) = 2\sigma(\xi_*) \sinh \Delta\xi$; their product expands to $\sigma(\xi_*)^2(e^{\Delta\xi} - 2 + e^{-\Delta\xi}) = 2\sigma(\xi_*)^2(\cosh \Delta\xi - 1)$. □

Integrated slopes above and below any fixed point pack into hyperbolic-trig functions of rapidity displacement. The curve carries a Lorentzian signature in $(\xi, \ln \sigma)$ coordinates.

4 Reciprocal-strike symmetry

Proposition 2 (Mode as geometric center). *Let ξ_C, ξ_P be rapidities with $\xi_C - \xi_m = -(\xi_P - \xi_m) =: \Delta \geq 0$. Then:*

- (i) *The slopes are reciprocals: $\sigma(\xi_C) \sigma(\xi_P) = 1$.*
- (ii) *The tangent angles satisfy $\tan \theta_C \tan \theta_P = \tan^2 \theta_m$: the mode is the geometric center of the curve in tangent-angle.*
- (iii) *The call premium at ξ_C equals the put premium at ξ_P , both equal to $e^{-\Delta}$, independently of w :*

$$P_C(\xi_C) = P_P(\xi_P) = e^{-\Delta}. \quad (15)$$

- (iv) *The same-side cross-premia are reciprocals: $P_C(\xi_C) P_C(\xi_P) = 1$, and $P_P(\xi_C) P_P(\xi_P) = 1$.*

Proof. (i) From (10), $\sigma(\xi_C) \sigma(\xi_P) = e^{(\xi_C + \xi_P) - 2\xi_m} = e^0 = 1$ since $\xi_C + \xi_P = 2\xi_m$. (ii) Exponentiate: $\tan \theta_C \tan \theta_P = e^{\xi_C + \xi_P} = e^{2\xi_m} = \tan^2 \theta_m$. (iii) $P_C(\xi_C) = e^{-(\xi_C - \xi_m)} = e^{-\Delta}$ and $P_P(\xi_P) = e^{\xi_P - \xi_m} = e^{-\Delta}$. (iv) $P_C(\xi_C) P_C(\xi_P) = e^{-(\xi_C - \xi_m) - (\xi_P - \xi_m)} = e^0 = 1$. □

Remark (The visible vs. invisible asymmetry). Reciprocal-strike pairs carry identical option premia at every w . Pool tilt ($w \neq 1/2$) therefore *cannot* produce asymmetry in cross-side premia. What it does produce is asymmetry in the *polar angular distances* $\Delta\theta_C := \theta_C - \theta_m$ and $\Delta\theta_P := \theta_m - \theta_P$, which are unequal whenever $w \neq 1/2$ even though the corresponding premia agree. The asymmetry lives in geometry, not in price.

5 Trade dynamics: the warp

5.1 Anchor and walk endpoint

The warp dynamics carry one fixed external datum beyond the running pool state: an **anchor point**

$$(x_0, y_0) \in V_0(\cdot) = k_0, \quad x_0, y_0 > 0, \quad (16)$$

chosen at pool deployment as any point on the initial curve. The anchor never moves, and—by construction below—it lies on every subsequent post-warp curve. A natural canonical choice is the mode point of the initial pool, but the formalization is anchor-agnostic: any point on the initial curve is admissible.

A trade with effective notional N at polar angle θ_* has **trade pivot**

$$(x_*, y_*) := (k (\tan \theta_*)^{w-1}, k (\tan \theta_*)^w) \in V(\cdot) = k. \quad (17)$$

Processing N against the curve via the side-specific slippage relation defines a **walk endpoint**

$$(x_B, y_B) \in V(\cdot) = k, \quad (18)$$

i.e. the point the trader would reach if the curve held fixed and reserves slid along it. The walk endpoint is a hypothetical reference, not a state update; it is the marginal-price target for the warp.

5.2 The warp rule

Definition (Curve warp). Let (x_*, y_*) be the trade pivot and (x_B, y_B) the walk endpoint, both on the pre-trade curve $V_0(\cdot) = k_0$. Define the **endpoint tangent**

$$\sigma_B := \frac{1 - w_0}{w_0} \cdot \frac{x_B}{y_B}, \quad (19)$$

i.e. the magnitude of dx/dy on the pre-trade curve at (x_B, y_B) . The post-trade state (w_1, k_1) is

$$w_1 := \frac{x_*}{x_* + \sigma_B y_*}, \quad (20)$$

$$k_1 := x_0^{w_1} y_0^{1-w_1}. \quad (21)$$

By (20), the new tangent slope σ at the trade ray θ_* on the post-trade curve equals σ_B : the warp installs the tangent the trader would have “walked into” on the pre-trade curve. By (21), the anchor lies on the new curve. The trade pivot generally does not: it slides radially along the ray θ_* to the new curve’s crossing.

5.3 Geometric reading

Proposition 3 (Warp is a rotation about the anchor in log-log). *Let $(u_0, v_0) := (\ln x_0, \ln y_0)$. The warp (20)–(21):*

- *fixes the anchor (u_0, v_0) on the line $wu + (1 - w)v = \ln k$;*
- *rotates the line about (u_0, v_0) , changing its slope from $-w_0/(1 - w_0)$ to $-w_1/(1 - w_1)$;*
- *slides the diagonal-crossing $(\ln k, \ln k)$ along the diagonal from $(\ln k_0, \ln k_0)$ to $(\ln k_1, \ln k_1)$;*
- *moves the ATM mode along a non-trivial trajectory: $(u_m^{(j)}, v_m^{(j)}) = (\ln k_j - (1 - w_j)\xi_m^{(j)}, \ln k_j + w_j\xi_m^{(j)})$.*

Proof. By (21), $V_1(x_0, y_0) = k_1$, so the anchor lies on the new curve, i.e. (u_0, v_0) is on the new line. By (20) together with (3), the tangent slope at any point of the new curve at polar angle θ_* is $-w_1/(1 - w_1) \cdot \tan \theta_* = -\sigma_B$. The diagonal-crossing is the line's intersection with $u = v$, giving $(\ln k_j, \ln k_j)$. The mode coordinates follow from (8). $\square$

Remark. The trade pivot (u_*, v_*) lies on the pre-trade line and on the polar ray θ_* . After the warp, the new line passes through (u_0, v_0) with a different slope, generally not through (u_*, v_*) . The pivot's new location is the intersection of the ray θ_* with the new line, at radial coordinate determined by k_1 .

5.4 Trade impact

Theorem 2 (Mode-rapidity shift equals negated walked rapidity). *With $\xi_* := \ln \tan \theta_* = \ln(y_*/x_*)$ and $\xi_B := \ln(y_B/x_B)$,*

$$\Delta\xi_m := \xi_m^{(1)} - \xi_m^{(0)} = \xi_* - \xi_B. \quad (22)$$

Proof. By (9), $\xi_m^{(j)} = \ln((1 - w_j)/w_j)$. From (20), $(1 - w_1)/w_1 = \sigma_B \cdot y_*/x_*$, hence $\xi_m^{(1)} = \ln \sigma_B + \xi_*$. From (19), $\ln \sigma_B = \ln((1 - w_0)/w_0) - \xi_B = \xi_m^{(0)} - \xi_B$. Substituting and rearranging gives (22). $\square$

The trader's walk along the pre-trade curve traverses rapidity displacement $\xi_B - \xi_*$. Theorem 2 states that the pool's mode rapidity shifts by exactly the negative of this displacement: *mode shift = minus walked rapidity*.

Corollary 3 (Closed form in trade notional). *For a call-side trade ($N = dy$ paid or received in the y -direction along the pre-trade curve), $y_B - y_* = \pm dy$ (sign: + for sell, - for buy), and using the curve invariant,*

$$\Delta\xi_m = \frac{1}{w_0} \ln \frac{y_*}{y_B}. \quad (23)$$

For a put-side trade ($N = dx$ in the x -direction), $\Delta\xi_m = \frac{1}{1 - w_0} \ln(x_B/x_)$, with the mirror sign convention.*

Proof. On the call side, the pre-trade curve gives $x_*^{w_0} y_*^{1-w_0} = x_B^{w_0} y_B^{1-w_0}$, hence $\ln(x_*/x_B) = ((1 - w_0)/w_0) \ln(y_B/y_*)$. Then $\xi_B - \xi_* = \ln(y_B/y_*) + \ln(x_*/x_B) = \ln(y_B/y_*)/w_0$. Apply Thm 2. The put side is symmetric. $\square$

Corollary 4 (Option-premium impact at the trade ray). *At the fixed rapidity ξ_* , the call and put premia rescale by reciprocal factors:*

$$\frac{P_C^{(1)}(\xi_*)}{P_C^{(0)}(\xi_*)} = e^{\Delta\xi_m}, \quad \frac{P_P^{(1)}(\xi_*)}{P_P^{(0)}(\xi_*)} = e^{-\Delta\xi_m}. \quad (24)$$

Proof. $P_C(\xi_*) = e^{\xi_m - \xi_*}$ scales as $e^{\Delta\xi_m}$; $P_P(\xi_*) = e^{\xi_* - \xi_m}$ scales as $e^{-\Delta\xi_m}$. $\square$

Corollary 5 (Slippage at the trade ray). *The change in side-specific option premium at the trade ray, pre- vs post-warp, is*

$$s_{\text{call}} = e^{\Delta\xi_m} - 1, \quad s_{\text{put}} = e^{-\Delta\xi_m} - 1. \quad (25)$$

A single scalar $\Delta\xi_m$ encodes the trade's full effect at the pivot.

5.5 Strike type-flip

Corollary 6 (Strike type-flip). *Let ξ_{strike} be the rapidity of a strike on the book. The strike's classification as a call (if $\xi_{\text{strike}} > \xi_m$) or put (if $\xi_{\text{strike}} < \xi_m$) is determined by the side of ξ_m on which it lies. Hence a trade with $\Delta\xi_m \neq 0$ flips the type of every strike whose rapidity lies strictly between $\xi_m^{(0)}$ and $\xi_m^{(1)}$:*

$$\min(\xi_m^{(0)}, \xi_m^{(1)}) < \xi_{\text{strike}} < \max(\xi_m^{(0)}, \xi_m^{(1)}) \implies \text{strike type flips.}$$

By Theorem 2, the swept band has size $|\Delta\xi_m| = |\xi_* - \xi_B|$, i.e. the rapidity gap between the trade ray and the walk-endpoint ray.

The strike's intrinsic position (ξ_{strike}) does not move; only its classification relative to the moving mode reference does. Strikes outside the swept band retain type and merely reprice per Corollary 4.

5.6 Visual summary

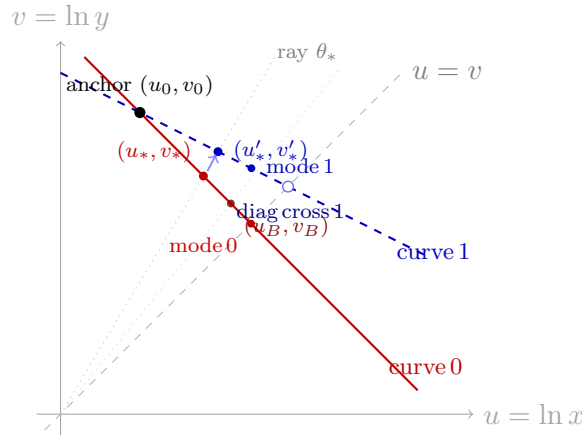


Figure: Model C warp in log-log (u, v) coordinates. The anchor (u_0, v_0) is fixed; the pricing line rotates about it (curve 0 $\rightarrow$ curve 1). The trade pivot $(u_*, v_*) \in$ curve 0 slides radially along the trade ray θ_* to $(u_*, v_*) \in$ curve 1. The walk endpoint $(u_B, v_B) \in$ curve 0 encodes the trader's rapidity displacement; Theorem 2 states $\Delta\xi_m = \xi_* - \xi_B$. For $w_0 = 0.5$ (symmetric initial pool), mode0 sits on the diagonal; mode1 for $w_1 = 1/3$ sits off the diagonal, distinct from the diagonal-crossing of curve 1.

Remark (On the anchor choice). If the anchor is taken at the initial pool's mode (in particular, on the diagonal when $w_0 = 1/2$), and the warp keeps the anchor on every subsequent curve, then the diagonal-crossing of every subsequent curve coincides with the anchor—i.e., $k_j = e^{u_0} = e^{v_0}$ for all j , so the pool scale k stays fixed across all trades. A generic anchor off the diagonal gives non-trivial k evolution.

6 Summary

Object	Identification
Pool curve	Straight line in $(\ln x, \ln y)$
Pool weight w	Line slope: $-w/(1-w)$
Pool scale k	Diagonal-crossing of line: $(\ln k, \ln k)$
ATM mode (off-diagonal for $w \neq 1/2$)	$(\ln k - (1-w)\xi_m, \ln k + w\xi_m)$
Rapidity ξ	$\ln \tan \theta = \ln y - \ln x$, intrinsic curve coord
Strike	Trade-invariant ξ_{strike} (or θ_{strike})
Mode rapidity ξ_m	$\ln((1-w)/w)$; moves under trade
Marginal slope	$\sigma(\xi) = e^{\xi - \xi_m}$
Call premium	$P_C(\xi) = e^{\xi_m - \xi}$
Put premium	$P_P(\xi) = e^{\xi - \xi_m}$
Premium duality	$P_C \cdot P_P \equiv 1$; both = 1 at mode (ATM)
Conservation (Thm 1)	$\sigma(\xi + \Delta\xi) \cdot \sigma(\xi - \Delta\xi) = \sigma(\xi)^2$
Reciprocal pair (Prop 2)	$\xi_C + \xi_P = 2\xi_m \Rightarrow P_C(\xi_C) = P_P(\xi_P)$
Anchor	Fixed point $(x_0, y_0) \in V_0(\cdot) = k_0$; on every post-warp curve
Walk endpoint (x_B, y_B)	Hypothetical pre-curve endpoint of the trade
Endpoint tangent σ_B	$(1-w_0)/w_0 \cdot x_B/y_B$ (target slope for warp)
Warp action	Rotation of line about anchor (u_0, v_0)
Pivot under warp	Slides radially along ray θ_* to new curve
Mode shift (Thm 2)	$\Delta\xi_m = \xi_* - \xi_B$ (negated walked rapidity)
Closed form (call/put)	$\Delta\xi_m = (1/w_0) \ln(y_*/y_B) / (1/(1-w_0)) \ln(x_B/x_*)$
Premium impact	P_C scales by $e^{\Delta\xi_m}$; P_P by $e^{-\Delta\xi_m}$
Slippage (call / put)	$e^{\Delta\xi_m} - 1 / e^{-\Delta\xi_m} - 1$
Strike type-flip (Cor 6)	$\xi_{\text{strike}} \in (\min(\xi_*, \xi_B), \max(\xi_*, \xi_B)) \Rightarrow$ type flips

The AMM is a one-dimensional rapidity object. Strikes are fixed points on this line; the mode is a moving cursor; trades shift the cursor by exactly minus the trader's walked rapidity on the pre-trade curve. Premia and slopes are evaluations at $\xi_{\text{strike}} - \xi_m$; the entire pool's response to a trade is one scalar.